

DATA VORTEX

Round 1 - Phase 1 | Data Intake Restoration

EXPLORATORY DATA ANALYSIS (EDA) REPORT

Dataset 01: Social Engine Users and Posts

Scope: Raw-data forensic audit, justified restoration of corrupted values, validation, and EDA of the recovered social-media-like dataset.

Analytical principle: Observe -> test -> validate -> decide -> transform -> validate again.

Submission context: The competition requires a cleaned dataset, EDA report, and documented/reproducible code workflow.

1. Executive Summary

Dataset 01 contains 12,360 post records and 1,500 user records. The Posts dataset contains several deliberate quality issues: missing values, 525 negative like counts, mixed timestamp representations, and exact duplicate source records. The cleaning process corrected invalid/missing values and standardized timestamps while preserving all rows.

The final analytical dataset contains 12,360 rows with no remaining missing values, no negative likes, no missing timestamps, and 12,360 unique physical-record identifiers. Source duplicate observations were preserved and explicitly flagged instead of being deleted.

Key EDA finding: Engagement is remarkably stable across many tested dimensions. Platform, weekday, language, location, text length, follower count, and hashtag/mention grouping did not show statistically significant differences/relationships in the tests performed.

Item	Result
Posts	12,360 rows x 8 raw columns
Users	1,500 rows x 5 raw columns
Final analytical Posts	12,360 rows x 10 columns
Missing values after cleaning	0
Negative likes after cleaning	0
Rows removed	0
Unique record_id values	12,360
Source duplicate-group rows preserved	712

2. Dataset Structure and Raw Audit

The raw Posts schema is: post_id, user_id, platform, text_content, timestamp, likes, shares, comments. The Users schema is: user_id, location, language, account_created, follower_count. Pandas initially inferred IDs/text/timestamps as object fields, likes as float64, and shares/comments/follower_count as integers.

A literal-token audit was performed with automatic NA interpretation disabled. This established how empty strings and literal NULL-like tokens were represented in the source file before standard Pandas NA handling.

Issue / field	Raw observation	Decision
platform missing	1,219 empty + 627 NULL = 1,846	Replace with Unknown
text_content missing	1,196 empty + 550 NULL + 24 NULL\n\n = 1,770	Replace with No text content
likes missing	1,229 empty + 629 NULL = 1,858	Median imputation = 2,500
negative likes	525 values	Sign restoration using absolute value
timestamps	3 valid representations	Standardize to datetime64[ns]
exact duplicates	360 extra duplicate copies; 352 duplicated post IDs	Preserve rows; flag duplicate groups

3. Forensic Reasoning Before Cleaning

Missingness was analyzed before imputation. Platform, text, and likes each had roughly 14-15% native missingness. Their overlap patterns were close to independence expectations, and chi-square tests found no statistically significant pairwise association between their missingness indicators.

Test	Statistic	p-value	Interpretation
Platform missing + text missing	chi2 = 2.1569	0.141928	No significant association
Platform missing + likes missing	chi2 = 0.7177	0.396916	No significant association
Text missing + likes missing	chi2 = 0.4263	0.513834	No significant association
Known platform vs missing likes	chi2 = 2.6434	0.619161	No significant association
Month vs missing likes	chi2 = 7.6038	0.748295	No significant association
User activity vs missing likes	chi2 = 0.9361	0.919330	No significant association

Negative likes were examined separately. Their absolute magnitudes fell within the normal 0-5,000 like range, and the absolute-value distribution was statistically consistent with the distribution of non-negative likes (KS statistic 0.0339, $p = 0.6007$). This supported sign restoration rather than setting every negative value to zero.

Correlations between likes and the other engagement components were negligible: likes-shares Pearson $r = 0.011846$ and likes-comments Pearson $r = 0.009988$. Therefore the missing-like values were not reconstructed from shares/comments.

4. Cleaning Decisions and Validation

Transformation	Execution	Validation / impact
Platform	Missing -> Unknown	0 missing; 12,360 rows retained
Text	Missing/sentinel -> No text content	0 missing; original observed text tracked separately
Likes sign correction	Negative likes -> absolute value	525 negatives -> 0
Likes imputation	Missing likes -> median 2,500	1,858 missing -> 0
Timestamp	Three validated formats -> datetime64[ns]	12,360 parsed; 0 failures
Duplicate handling	No deletion; record_id + duplicate flag	12,360 rows; 12,360 unique record IDs

The final export was reloaded and independently checked. Shape was (12,360, 10), missing counts were zero, negative likes were zero, and duplicate record_id values were zero.

5. Exploratory Data Analysis

5.1 Numerical engagement profile

Metric	Mean	Median	Std. dev.	Min	Max
Likes	2493.84	2500	1324.88	0	5000
Shares	1005.87	1016	574.73	0	2000
Comments	504.08	503	288.80	0	1000

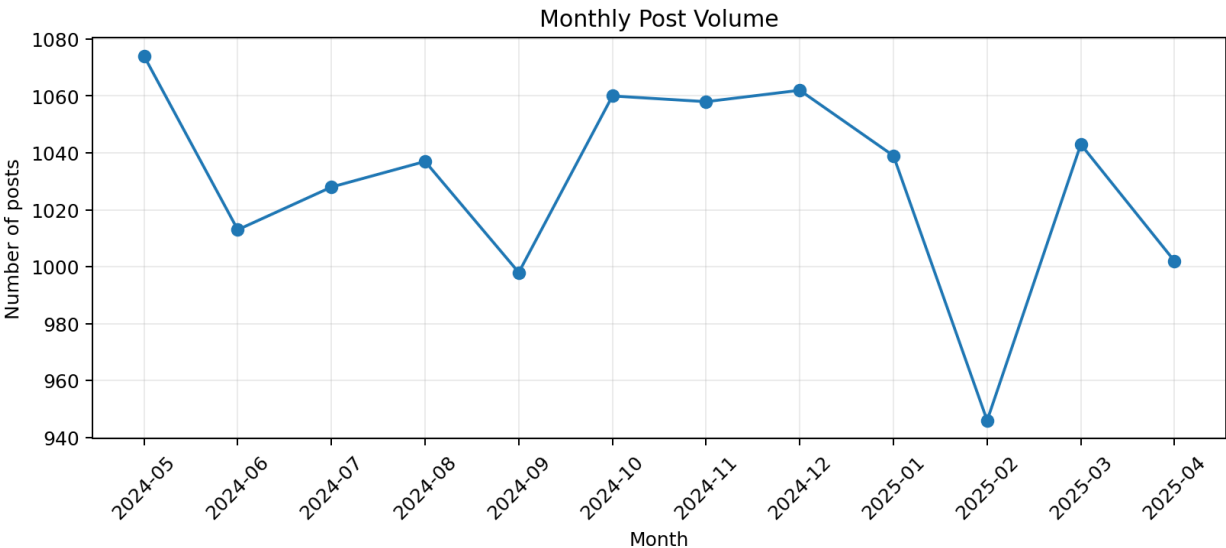
The cleaned engagement distributions are approximately symmetric by skewness: likes 0.0068, shares -0.0118, comments -0.0156. No IQR outliers were detected for likes, shares, comments, or total composite engagement.

5.2 Platform performance

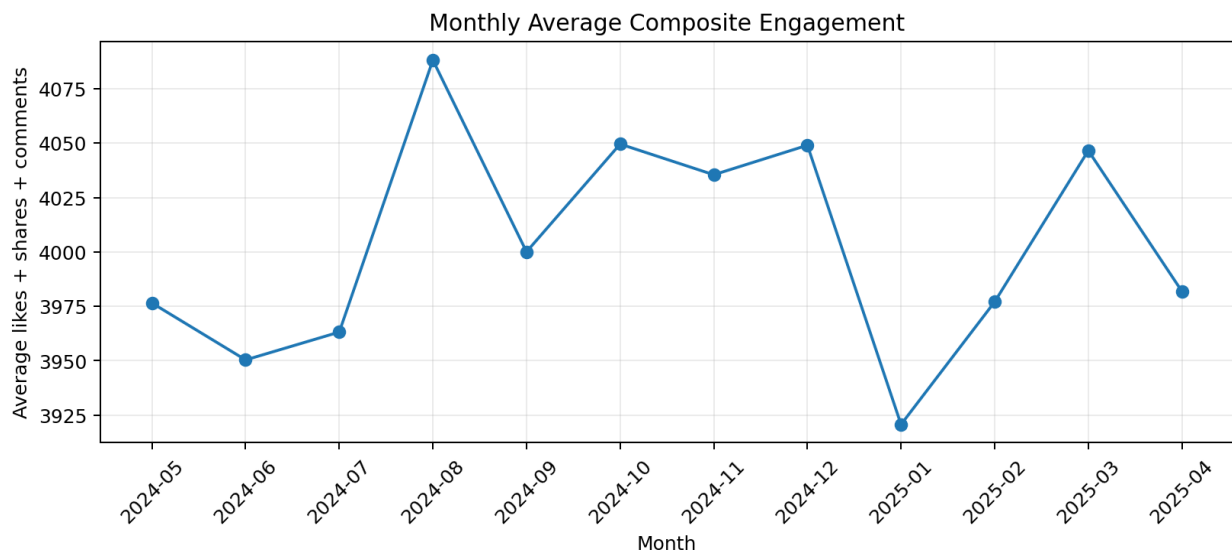
Platform	Posts	Avg total engagement	Engagement share
Instagram	2,038	4,044.22	16.66%
YouTube	2,136	4,031.49	17.40%
Facebook	2,135	4,013.49	17.32%
Reddit	2,086	3,999.77	16.86%
Unknown	1,846	3,980.92	14.85%
Twitter	2,119	3,951.10	16.92%

Instagram has the highest descriptive average composite engagement per post (4,044.22). YouTube contributes the highest total engagement (8,611,269; 17.40% of the composite total). Platform engagement share is close to platform post share, so no platform shows a dramatic aggregate advantage.

5.3 Temporal activity



May 2024 is the busiest month at 1,074 posts; February 2025 is the lowest at 946. There is no sustained upward or downward posting trend over the one-year observation period.



Average composite engagement peaks in August 2024 at 4,088.19 and is lowest in January 2025 at 3,920.71. These are descriptive peaks; one year of data is insufficient to claim seasonality.

6. Content and User Behaviour

6.1 Weekday activity

Posting volume is broadly uniform across the seven days. A chi-square test gave $\chi^2 = 4.1487$, $p = 0.65656$. One-way ANOVA of average composite engagement by weekday gave $F = 0.8016$, $p = 0.568456$. Thus, weekday differences are not statistically significant.

6.2 Text length

Among 10,590 posts with genuine observed text, mean length is 117.63 characters and median length is 118. Spearman correlation between text length and composite engagement is $\rho = -0.0138$, indicating a negligible relationship.

6.3 Hashtags and mentions

Among genuine-text posts, 10,520 contain hashtags, 1,415 contain mentions, and 70 contain neither. The structure is: 9,105 hashtag-only, 1,415 hashtag+mention, 70 neither. No mention-only posts occur. One-way ANOVA across these content groups gave $F = 1.3351$, $p = 0.263166$, so the descriptive differences are not statistically significant.

6.4 Follower count, location, and language

Analysis	Result	Conclusion
Follower count vs avg engagement	Pearson $r = -0.0017$; Spearman $\rho = 0.0047$	No meaningful relationship
Location vs avg engagement	$F = 0.7559$; $p = 0.835694$	No significant location effect
Language vs avg engagement	$F = 1.1293$; $p = 0.338218$	No significant language effect

Location groups contain 28-56 users each; language groups contain 141-168 users each. Delhi has the highest descriptive location mean (4,098.48) and Vancouver the lowest (3,823.60), but the location ANOVA does not support a statistically significant difference. English has the highest descriptive language mean (4,071.26) and Hindi the lowest (3,896.21), again without a significant overall language effect.

7. Engagement Composition and Limitations

In the cleaned dataset, the defined composite engagement is likes + shares + comments. Its average composition is 58.71% likes, 26.96% shares, and 14.33% comments. A robustness check using only originally valid likes gives 57.76%, 27.62%, and 14.62%, respectively. The differences are modest, but likes proportions remain partly influenced by the 1,858 median imputations.

Key limitations: (1) 1,858 likes were imputed with the observed median of 2,500; (2) 1,770 text entries were represented as missing/no text and excluded from text-content analyses; (3) 360 extra duplicate copies were retained because no-row-removal was required; (4) composite engagement is a derived metric, not an organizer-provided field; and (5) inferential tests are descriptive evidence within this dataset and do not establish causality.

8. Final Insights for Judges

1. **Engagement is stable across many dimensions.** Platform, weekday, location, language, text length, follower count, and hashtag/mention grouping do not show strong statistically significant effects in the tests performed.
2. **Volume and intensity are different.** May 2024 has the highest posting volume, while August 2024 has the highest average engagement per post. The busiest month is therefore not the most engaging month.
3. **Platform leadership depends on the metric.** Instagram leads descriptive average engagement per post, while YouTube leads aggregate engagement contribution.
4. **Missing likes were not arbitrarily reconstructed.** Forensic testing supported sign restoration for negative values, while missing likes were handled with robust global median imputation because no tested platform/time/user-activity factor explained their missingness and shares/comments had negligible linear association with likes.
5. **No engagement outliers were detected under the 1.5 x IQR rule.** High-engagement posts remain valid observations rather than cleaning targets.

End of Phase 1 EDA Report